

EXPERIMENT REPORT

Design and Measurement of a
Multistage Amplifier with Feedback

SPICE Simulation, Hardware Testing, and Performance Analysis

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Theme color: purple. The report is based on the uploaded instruction manual and the uploaded simulation/measurement screenshots. All supplied result screenshots are included as figures.

1. Abstract

This report presents the design, simulation, hardware implementation, and performance measurement of a two-stage MOSFET multistage amplifier with feedback. The circuit was designed in LTspice using two 2N7002 MOSFETs, a 12 V DC supply, AC coupling capacitors, bias resistors, source degeneration resistors, and a 100 k ohm feedback resistor. The experiment followed the instruction sequence of schematic construction, DC operating-point analysis, transient simulation, small-signal AC analysis, and input/output resistance measurement. The supplied experimental screenshots are incorporated throughout the report as evidence of the simulation and measurement process.

The LTspice results show that both MOSFET stages are biased at approximately 1.2 mA and that the amplifier provides a midband gain of about 24 dB with a sinusoidal output in the transient test. Hardware oscilloscope screenshots confirm sinusoidal operation at 1 kHz and 10 kHz, while the amplitude decreases at 100 kHz because of high-frequency rolloff. From the two-channel resistance measurements, the input resistance is approximately 49 k ohm and the output resistance is approximately 7.8 k ohm, under the assumptions stated in the calculation section.

2. Objectives

- Build and simulate a two-stage amplifier with feedback in LTspice.
- Verify the DC bias point so that both MOSFETs operate in a suitable amplification region.
- Observe the time-domain input and output waveforms under a sinusoidal input signal.
- Use small-signal AC analysis to estimate midband gain, passband, and cutoff behavior.
- Measure input resistance and output resistance using the voltage-divider and loaded-output methods.
- Compare the simulated behavior with the breadboard and oscilloscope measurements.

3. Experimental Principle

The amplifier is composed of two AC-coupled MOSFET stages using 2N7002 devices. The first stage establishes the main voltage gain, and the second stage provides additional amplification and output driving capability. Resistor dividers set the MOSFET gate bias, source resistors stabilize drain current, and the coupling capacitors block DC while allowing the AC signal to pass. The 100 k ohm feedback path stabilizes the operating point and helps reduce sensitivity to device-parameter variation.

$$\text{Voltage gain: } A_v = V_o / V_i, \quad A_v(\text{dB}) = 20 \log_{10}(|A_v|)$$

$$\text{Input resistance: } R_{in} = R_s * V_i / (V_s - V_i)$$

$$\text{Output resistance: } R_{out} = R_L * (V_o / V_L - 1)$$

For the input-resistance method, a known series resistance R_s is inserted between the signal source and the amplifier input. The source-side voltage V_s and the amplifier input voltage V_i are compared. For the output-resistance method, the no-load output voltage V_o is compared with the loaded output voltage V_L when a known load R_L is connected.

4. Circuit Design

The LTspice schematic uses a 12 V supply, two 2N7002 MOSFETs, a 1 k ohm source-series input resistor, 1 uF coupling capacitors, a 10 k ohm load, and the following main resistors: $R_5 = 240$ k ohm, $R_2 = 118$ k ohm, $R_6 = 7.3$ k ohm, $R_7 = 8.5$ k ohm, $R_3 = 1.8$ k ohm, $R_4 = 1$ k ohm, and $R_8 = 100$ k ohm. The input source is a 1 kHz sinusoid with 10 mV amplitude in the simulation screenshot.

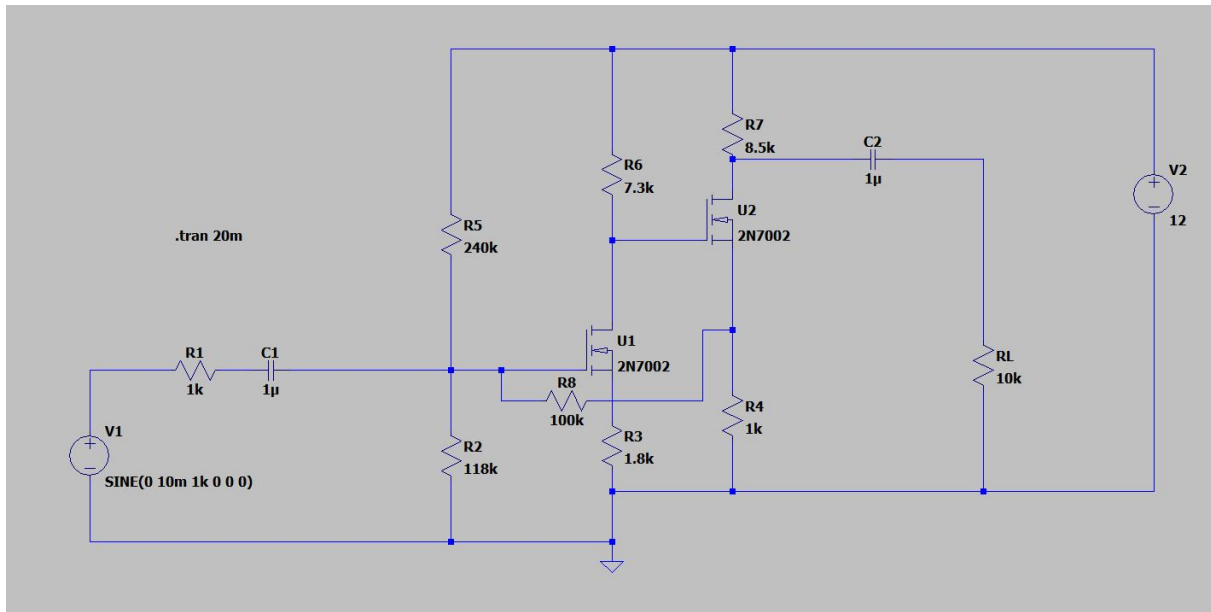


Figure 1. LTspice schematic of the two-stage 2N7002 amplifier with feedback.

5. Procedure

The procedure follows the uploaded instruction manual. First, the amplifier schematic is established in LTspice. Second, a DC operating-point command is run to verify the bias voltages and device currents. Third, a transient command is used to observe the input and output sinusoidal waveforms. Fourth, a small-signal AC sweep is performed to obtain the gain and phase response versus frequency. Finally, input resistance and output resistance are measured by comparing voltages with known series or load resistors.

Step	Operation	Purpose
1	Schematic construction	Build the two-stage amplifier and define component values.
2	DC operating point	Run .op to check node voltages and MOSFET currents.
3	Transient analysis	Apply the sinusoidal source and run a 20 ms transient simulation.
4	AC analysis	Set a small-signal AC source and sweep frequency to find gain and bandwidth.
5	Resistance measurements	Use V_s/V_i for input resistance and V_o/V_L for output resistance.

6. Simulation Results

6.1 DC Operating Point

The DC operating-point output shows that the supply is 12 V and both MOSFETs conduct approximately 1.2 mA. This confirms that the amplifier has a stable bias condition before the AC and transient simulations are performed.

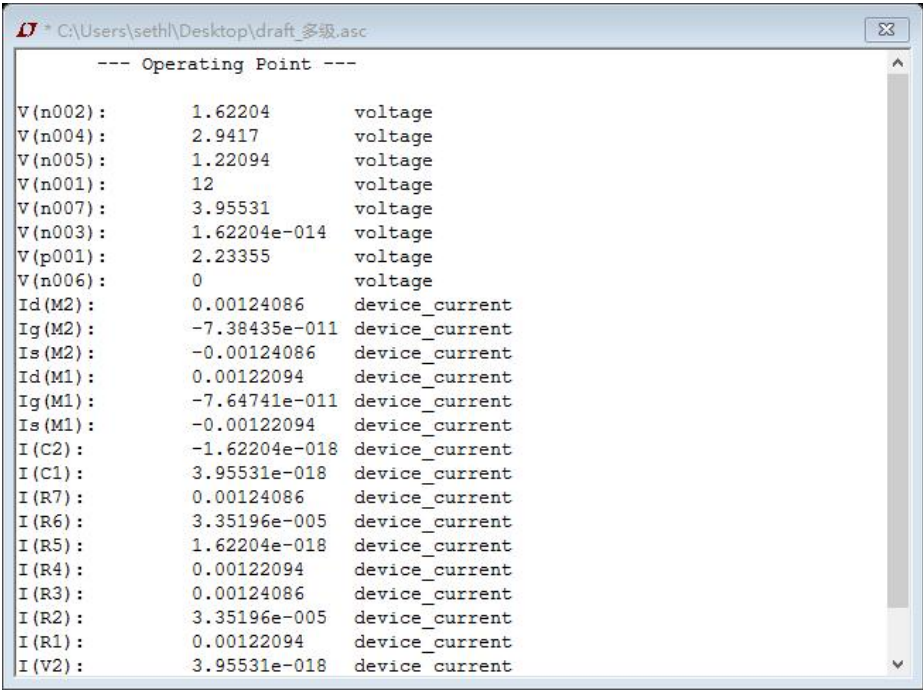


Figure 2. LTspice DC operating-point result.

LTspice item	Value	Comment
V(n002)	1.62204 V	Bias node in the first-stage input network
V(n004)	2.9417 V	Internal bias/output node shown in LTspice
V(n005)	1.22094 V	Source-related bias node
V(n007)	3.95531 V	Second-stage bias node
Id(M1)	1.22094 mA	Drain current of M1
Id(M2)	1.24086 mA	Drain current of M2

6.2 Transient Analysis

The transient waveform is sinusoidal and does not show obvious clipping. The input is about 20 mV peak-to-peak, while the output is about 340 mV peak-to-peak. Therefore, the transient voltage gain is approximately 17 V/V, which corresponds to about 24.6 dB.

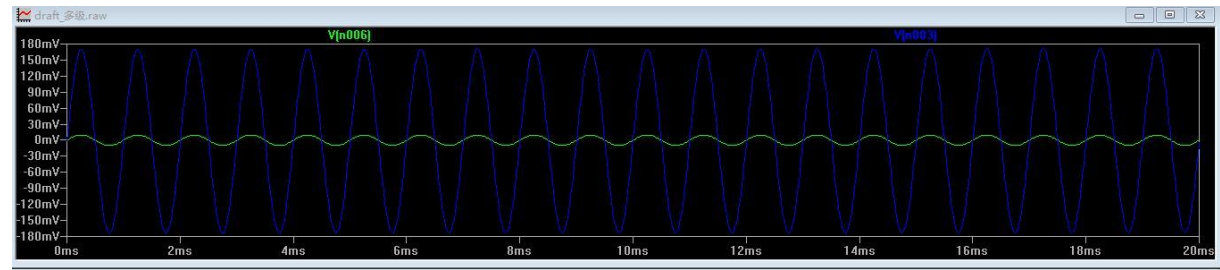


Figure 3. Transient simulation showing input and amplified output waveforms.

6.3 Small-Signal AC Analysis

The AC sweep shows a midband magnitude close to 24 dB. The low-frequency gain rises as the coupling capacitors become effective, and the high-frequency gain rolls off because of device capacitances and circuit parasitics. Based on the screenshot, the lower cutoff is around 10 Hz and the upper cutoff is in the tens of kilohertz range. The passband therefore covers the 1 kHz and 10 kHz hardware test frequencies, while 100 kHz lies near or beyond the high-frequency rolloff region.

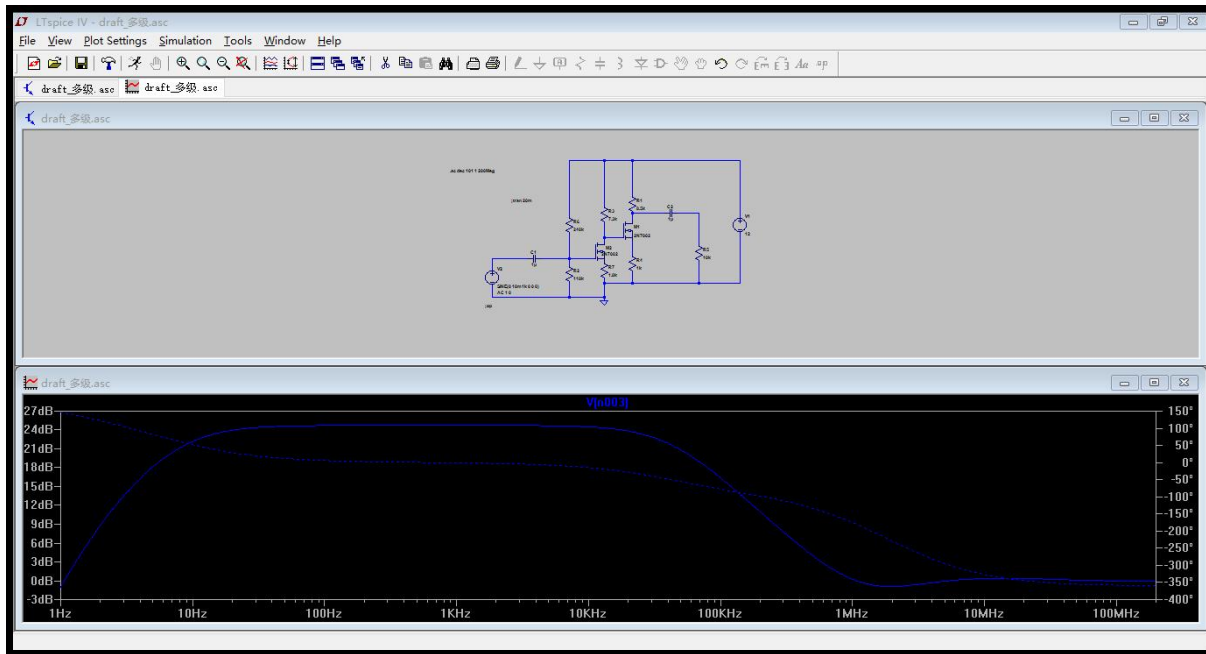


Figure 4. LTspice AC response; solid trace is gain and dashed trace is phase.

6.4 Simulated Input and Output Resistance Tests

The resistance test screenshots show the voltage-comparison method required by the instruction manual. For output resistance, the difference between no-load and loaded output amplitudes indicates the internal Thevenin resistance of the amplifier output. For input resistance, the close overlap of the source-side and input-side waveforms indicates a relatively high input resistance compared with the series test resistor.

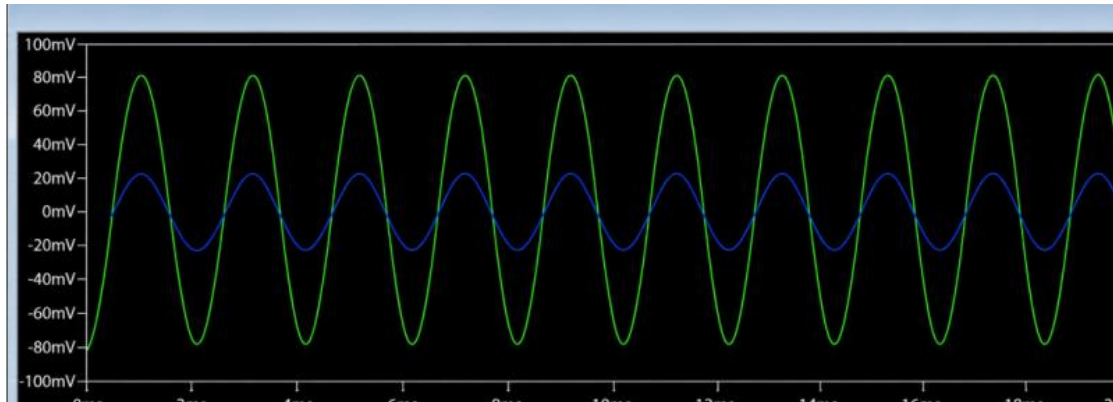


Figure 5. Simulation waveform used for output-resistance evaluation.

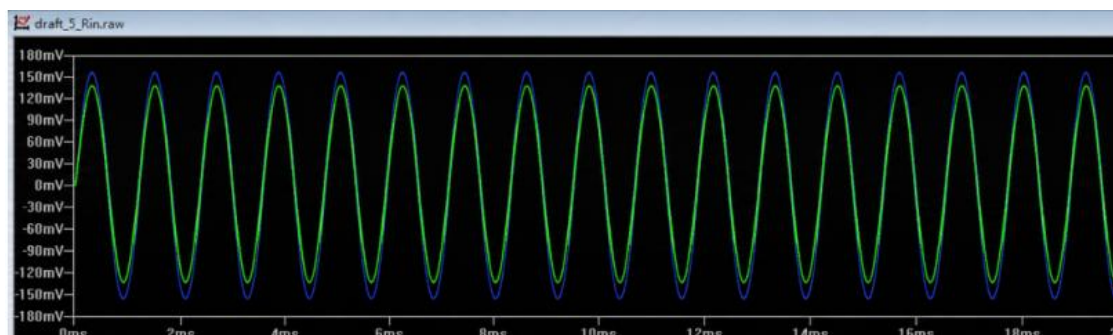


Figure 6. Simulation waveform used for input-resistance evaluation.

7. Hardware Measurement Results

After simulation, the amplifier was assembled on a breadboard. The oscilloscope photographs show the measured output waveforms at different frequencies and the two-channel measurements used for input and output resistance calculations.

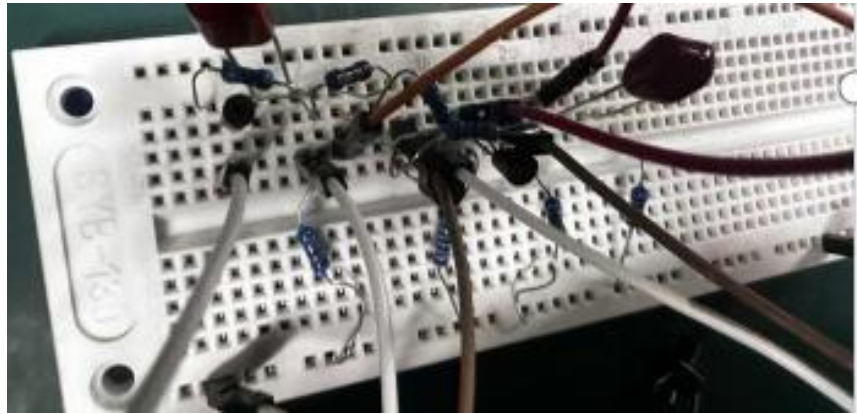


Figure 7. Breadboard implementation of the multistage amplifier.

7.1 Frequency-Domain Behavior in Hardware

The oscilloscope readings at 1 kHz and 10 kHz remain close in amplitude, which agrees with the simulated passband behavior. At 100 kHz, the output amplitude is much smaller, confirming the high-frequency attenuation predicted by AC analysis.

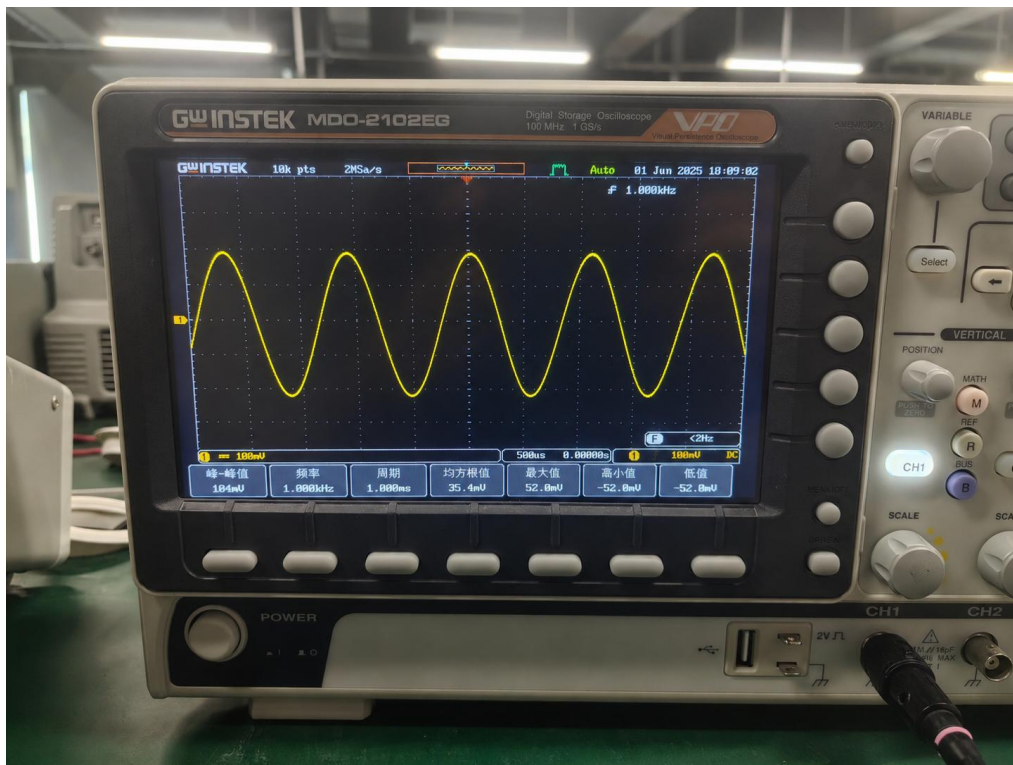


Figure 8. Hardware output waveform at 1 kHz; the oscilloscope displays about 104 mV peak-to-peak.

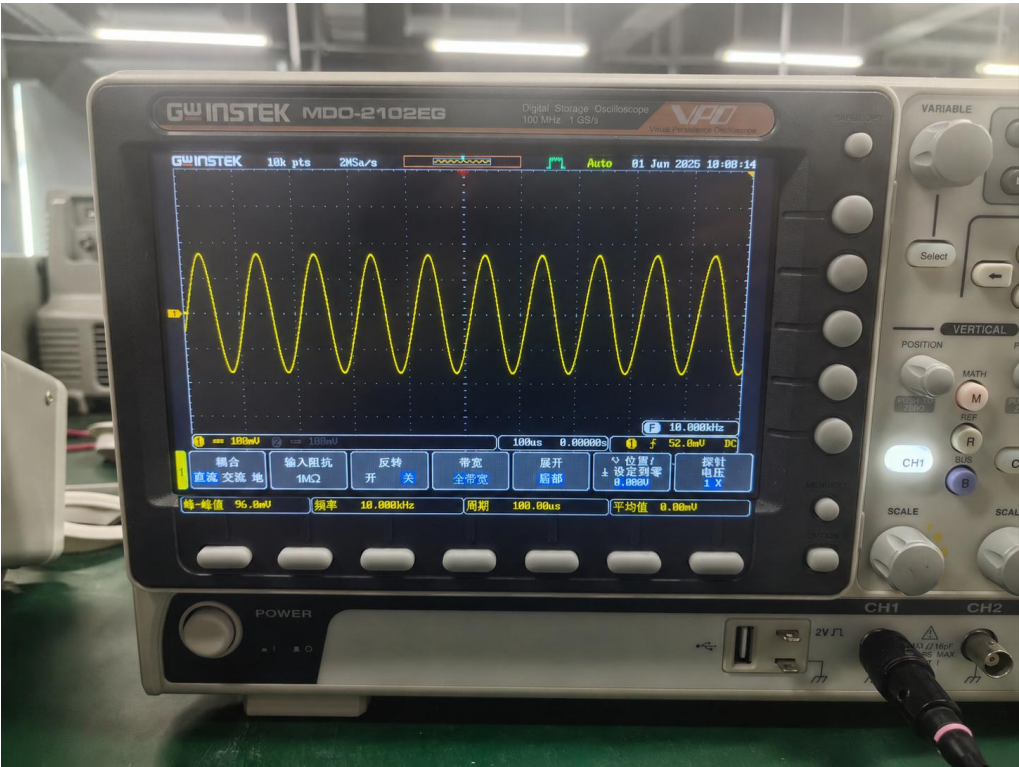


Figure 9. Hardware output waveform at 10 kHz; the oscilloscope displays about 96.0 mV peak-to-peak.

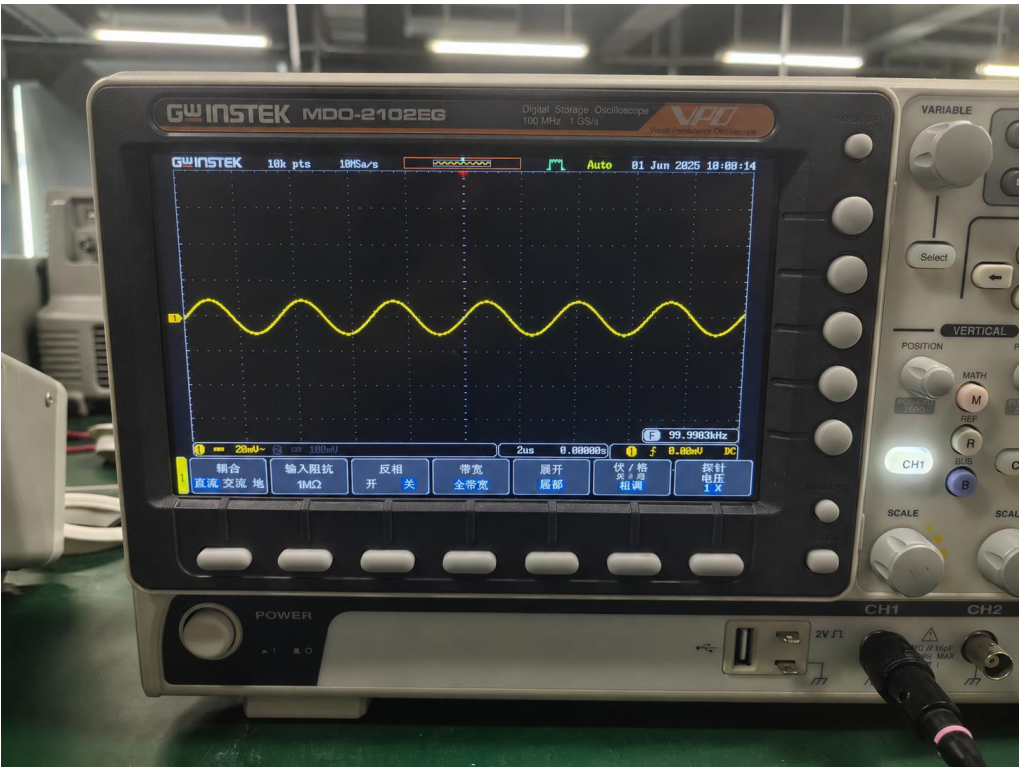


Figure 10. Hardware output waveform at approximately 100 kHz; the amplitude is clearly reduced.

Test	Displayed frequency	Output amplitude	Observation
1 kHz	1.000 kHz	104 mV p-p	Clean sine wave in passband
10 kHz	10.000 kHz	96.0 mV p-p	Amplitude remains close to 1 kHz result
100 kHz	99.998 kHz	Approximately 30-40 mV p-p from screen grid	High-frequency rolloff is evident

7.2 Input Resistance Measurement

In the input-resistance screenshot, the two oscilloscope channels are almost overlapped at 1 kHz. The displayed peak-to-peak voltages are approximately 20.0 mV for the source-side signal and 19.6 mV for the amplifier input. Assuming the known series resistor is $R_s = 1 \text{ k}\Omega$, the input resistance is:

$$R_{in} = 1 \text{ k}\Omega * 19.6 \text{ mV} / (20.0 \text{ mV} - 19.6 \text{ mV}) = 49 \text{ k}\Omega$$

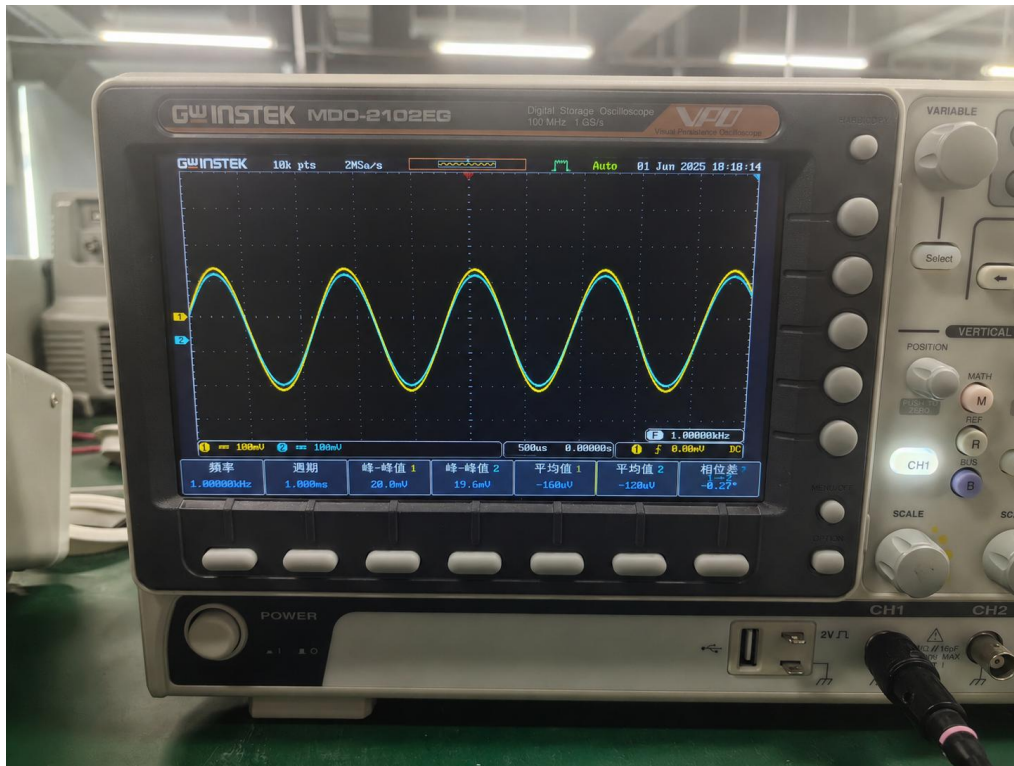


Figure 11. Hardware input-resistance measurement; CH1 is about 20.0 mV p-p and CH2 about 19.6 mV p-p.

7.3 Output Resistance Measurement

In the output-resistance screenshot, the two measured waveforms at 1 kHz have different amplitudes. Taking CH1 as the no-load output voltage and CH2 as the loaded output voltage, the displayed values are approximately $V_o = 160 \text{ mV p-p}$ and $V_L = 90 \text{ mV p-p}$. Assuming the load resistor is $R_L = 10 \text{ k}\Omega$, the output resistance is:

$$R_{out} = 10 \text{ k}\Omega * (160 \text{ mV} / 90 \text{ mV} - 1) = 7.8 \text{ k}\Omega$$

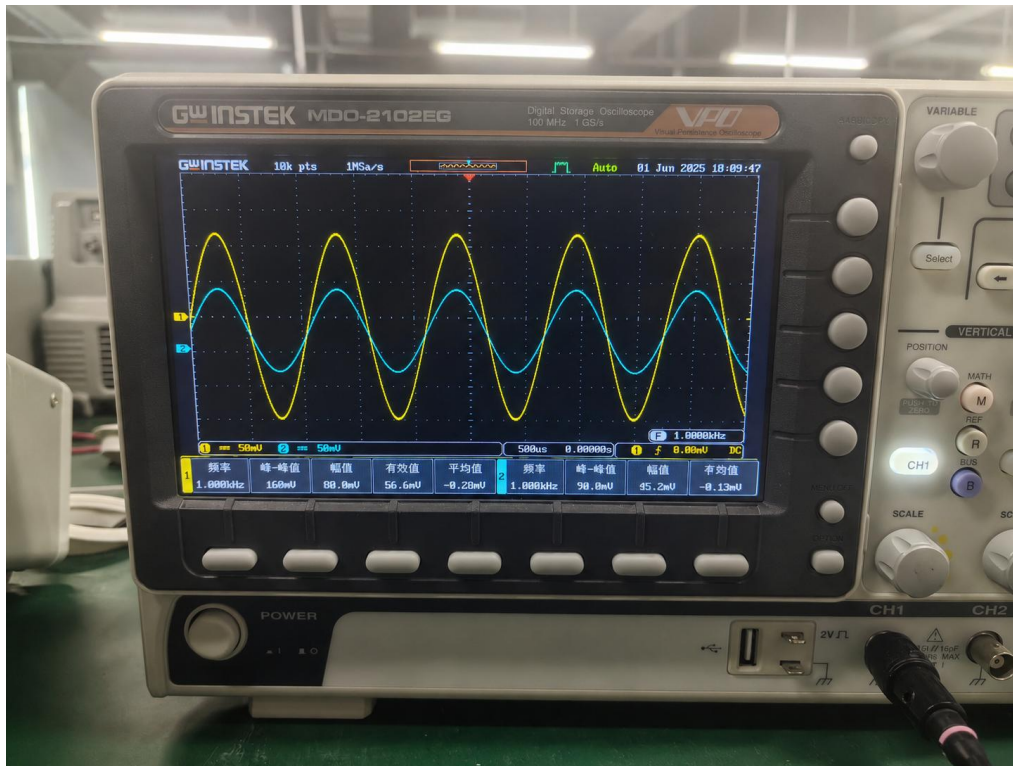


Figure 12. Hardware output-resistance measurement; CH1 is about 160 mV p-p and CH2 about 90 mV p-p.

8. Discussion

The simulation and measurement results are consistent in their main trends. The amplifier has a stable DC operating point, provides useful gain in the midband, and exhibits high-frequency rolloff. The AC analysis explains why the hardware output remains large at 1 kHz and 10 kHz but becomes much smaller at 100 kHz.

The measured input resistance is relatively high because the MOSFET gate draws very little DC current and the bias network dominates the input loading. The measured output resistance is close to the order of the second-stage drain resistance, which is expected for a common-source output node with a finite drain-load resistor. Differences between simulation and hardware can be caused by MOSFET threshold-voltage spread, resistor and capacitor tolerance, breadboard parasitic capacitance, probe loading, and imperfect signal-source calibration.

Assumptions for calculated resistance values: The screenshot document does not explicitly state the external test resistor values. The calculations use $R_s = 1 \text{ k } \Omega$ from the input series resistor shown in the schematic and $R_L = 10 \text{ k } \Omega$ from the load resistor shown in the schematic. If different external test resistors were used in the laboratory, R_{in} and R_{out} should be recalculated with those values.

9. Conclusion

The multistage amplifier with feedback was successfully designed, simulated, and measured. The LTspice DC operating point confirms that both 2N7002 MOSFETs are biased with drain currents of approximately 1.2 mA. The transient simulation gives a sinusoidal amplified output with an estimated gain of about 17 V/V, and the AC sweep gives a midband gain of about 24 dB. Hardware measurements show a clear output waveform at 1 kHz and 10 kHz and a reduced output amplitude at 100 kHz, matching the expected frequency-response rolloff. The estimated input resistance is about 49 k Ω and the output resistance is about 7.8 k Ω based on the uploaded oscilloscope screenshots and the stated assumptions. Overall, the experiment demonstrates the complete workflow for designing a multistage amplifier in SPICE and validating its performance in hardware.

10. References

No.	Source
[1]	Uploaded experimental instruction manual: "实验 4 多级放大器_反馈.pptx".
[2]	Uploaded simulation and measurement screenshot document: "实验四仿真+测量.docx".